```
!pip install tensorflow

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt
import numpy as np

print("TensorFlow version:", tf.__version__)
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
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Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboar
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tens
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3
TensorFlow version: 2.19.0
```

```
# Load dataset
(X_train, y_train), (X_test, y_test) = keras.datasets.mnist.load_data()

# Normalize pixel values (0–255 → 0–1)
X_train = X_train / 255.0
X_test = X_test / 255.0

# Reshape to CNN format (samples, height, width, channels)
X_train = X_train.reshape(-1, 28, 28, 1)
X_test = X_test.reshape(-1, 28, 28, 1)

print("Training shape:", X_train.shape)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Training shape: (60000, 28, 28, 1)
```

```
model = keras.Sequential([

    layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),

    layers.Dense(64, activation='relu'),
```

```
    layers.Dense(10, activation='softmax')
])

model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shap
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 64)	102,464
dense_1 (Dense)	(None, 10)	650

Total params: 121,930 (476.29 KB)
Trainable params: 121,930 (476.29 KB)
Non-trainable params: 0 (0.00 B)

```
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

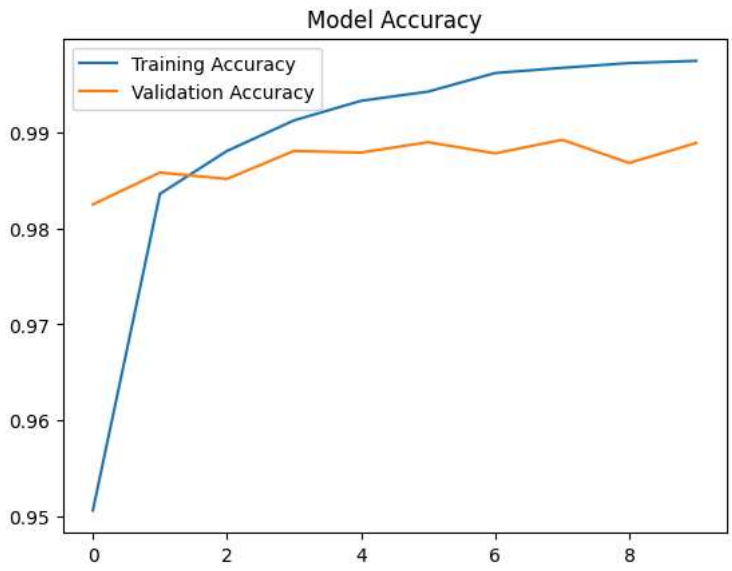
```
history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=32,
    validation_split=0.2
)
```

```
Epoch 1/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 43s 28ms/step - accuracy: 0.8898 - loss: 0.3601 - val_accuracy: 0.9825 - val_loss: 0.0659
Epoch 2/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 41s 27ms/step - accuracy: 0.9824 - loss: 0.0566 - val_accuracy: 0.9858 - val_loss: 0.0486
Epoch 3/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 40s 27ms/step - accuracy: 0.9886 - loss: 0.0341 - val_accuracy: 0.9852 - val_loss: 0.0526
Epoch 4/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 41s 27ms/step - accuracy: 0.9912 - loss: 0.0270 - val_accuracy: 0.9881 - val_loss: 0.0399
Epoch 5/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 41s 27ms/step - accuracy: 0.9933 - loss: 0.0192 - val_accuracy: 0.9879 - val_loss: 0.0410
Epoch 6/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 40s 26ms/step - accuracy: 0.9953 - loss: 0.0140 - val_accuracy: 0.9890 - val_loss: 0.0388
Epoch 7/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 42s 27ms/step - accuracy: 0.9968 - loss: 0.0103 - val_accuracy: 0.9878 - val_loss: 0.0457
Epoch 8/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 44s 29ms/step - accuracy: 0.9966 - loss: 0.0093 - val_accuracy: 0.9893 - val_loss: 0.0489
Epoch 9/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 41s 27ms/step - accuracy: 0.9978 - loss: 0.0071 - val_accuracy: 0.9868 - val_loss: 0.0542
Epoch 10/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 40s 26ms/step - accuracy: 0.9975 - loss: 0.0076 - val_accuracy: 0.9889 - val_loss: 0.0481
```

```
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", test_acc)
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 4s 11ms/step - accuracy: 0.9879 - loss: 0.0478
Test Accuracy: 0.989799976348877
```

```
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.title("Model Accuracy")
plt.show()
```


Model Accuracy
Training Accuracy
Validation Accuracy
0.99
0.98
0.97
0.96
0.95
0
2
4
6
8